



Kullback–Leibler average, consensus on probability densities, and distributed state estimation with guaranteed stability[☆]



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ABSTRACT

This paper addresses distributed state estimation over a sensor network wherein each node – equipped with processing, communication and sensing capabilities – repeatedly fuses local information with information from the neighbors. Estimation is cast in a Bayesian framework and an information-theoretic approach to data fusion is adopted by formulating a consensus problem on the Kullback–Leibler average of the local probability density functions (PDFs) to be fused. Exploiting such a consensus on local posterior PDFs, a novel distributed state estimator is derived. It is shown that, for a linear system, the proposed estimator guarantees stability, i.e. mean-square boundedness of the state estimation error in all network nodes, under the minimal requirements of network connectivity and system observability, and for any number of consensus steps. Finally, simulation experiments demonstrate the validity of the proposed approach.

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1. Introduction

Nowadays multi-agent systems, i.e. networks consisting of multiple nodes (agents) with local data acquisition, processing and communication capabilities, are ubiquitous in automation and supervision systems in both industrial and other contexts (e.g., environmental monitoring). Such systems pose interesting problems that need to be thoroughly investigated from both theoretical and practical viewpoints. In this respect, a challenging research topic is to devise scalable and robust techniques for distributed information fusion in multi-agent systems. A still open issue is how to effectively counteract the deleterious effects of “data incest” that occur even in the simple case in which multiple inter-communicating agents, without any centralized coordination and according to an unknown and possibly time-varying network topology, try to estimate the state of a linear dynamical system; in fact, in such a case, it is common to find, due to the correlation among the estimates of different agents, a remarkable performance degradation with respect to the centralized Kalman estimator or even, if the target process is unstable, divergence of the estimation error. A key objective

to be pursued is, therefore, to develop, for the state estimation of a linear process, distributed state estimation algorithms (to be implemented in each agent) that guarantee for each agent, under the minimal assumptions of process observability from the whole network (but not necessarily from the individual agents) and strong network connectivity, bounded estimation error as close as possible to the one obtained with the optimal centralized estimator. Among the approaches devised to counteract the data incest phenomenon, special attention is deserved by the consensus filters (Olfati-Saber, Fax, & Murray, 2007; Xiao, Boyd, & Lall, 2005) and by the covariance intersection fusion rule (Chen, Arambel, & Mehra, 2002; Julier & Uhlmann, 1997). Consensus filters provide a general tool for distributed averaging; in the context of distributed state estimation, they have been exploited for averaging innovations (Kamgarpour & Tomlin, 2008; Olfati-Saber, 2007) or state estimates (Alriksson & Rantzer, 2006; Carli, Chiuso, Schenato, & Zampieri, 2008; Olfati-Saber, 2009; Stankovic, Stankovic, & Stipanovic, 2009). The covariance intersection fusion adopts an information filtering approach, by propagating the information (inverse covariance) matrix and the information vector defined via pre-multiplication of the state estimate by the information matrix, and computes the fused information pair as a convex combination of the local information pairs. Other relevant approaches to distributed state estimation include those based on moving-horizon estimation (Farina, Ferrari-Trecate, & Scattolini, 2010) and diffusion processes (Cattivelli & Sayed, 2010).

In this paper, a general Bayesian approach to distributed state estimation will be adopted by formulating a consensus problem on the agent probability density functions (PDFs) of the state vector to be estimated and an information-theoretic perspective

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will be provided by defining the average PDF as the one that minimizes the sum of the information gains (Kullback–Leibler divergences) from the original (agent) PDFs. In this respect, it will be shown that the Kullback–Leibler average PDF turns out to be the geometric mean of the agent PDFs. It will also be shown that, for Gaussian PDFs and a single consensus step, consensus on PDFs reduces to the well known covariance intersection fusion. This suggests how multistep consensus can be exploited in order to possibly improve performance of the covariance intersection fusion. Another contribution of this paper will be to prove that, for linear systems, distributed state estimation with consensus on the posterior PDFs guarantees, under minimal requirements on network connectivity and system observability as well as for any number of consensus steps, bounded estimation error covariance in all network nodes.

The rest of the paper is organized as follows. Section 2 formulates consensus on PDFs in terms of Kullback–Leibler averaging and then shows how this reduces to covariance intersection for Gaussian PDFs and a single consensus step. Next, Section 3 applies this approach to distributed state estimation of a discrete-time dynamical system and derives a novel algorithm based on the consensus among the local posterior PDFs. In Section 4, stability analysis of the proposed distributed state estimator is carried out in the linear system case. Then, Section 5 evaluates the performance of such an estimator by simulation experiments. Finally, Section 6 ends the paper with concluding remarks and perspectives for future work.

2. Consensus on the Kullback–Leibler average

Consider a network consisting of N nodes. Using graph formalism, the communication structure among the nodes is represented by a directed graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ where $\mathcal{N} := \{1, 2, \dots, N\}$ is the set of nodes and $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ is the set of links. In particular, it is supposed that (j, i) belongs to $\mathcal{E}$ if and only if node i can receive data from node j (by definition $(i, i) \in \mathcal{E}$ for any $i \in \mathcal{N}$). Further, let $\mathcal{N}^i = \{j \in \mathcal{N} : (j, i) \in \mathcal{E}\}$ denote the set of in-neighbors of node i (note that, by definition, i always belongs to $\mathcal{N}^i$). Finally, let $|\mathcal{N}^i|$ be the cardinality of $\mathcal{N}^i$ (i.e., the in-degree of node i plus one).

Let

$$\mathcal{P} \triangleq \left\{ p(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R} \text{ such that} \right. \\ \left. \int_{\mathbb{R}^n} p(x) dx = 1 \text{ and } p(x) \geq 0, \forall x \in \mathbb{R}^n \right\}$$

denote the set of PDFs over $\mathbb{R}^n$. Suppose that, in each node i of the network, a PDF $p^i(\cdot) \in \mathcal{P}$ is available representing the local information on some random vector $x \in \mathbb{R}^n$ of interest. For example, such PDFs can be obtained via statistical inference or can be the result of some recursive Bayesian estimation algorithm (as will be discussed in the following sections).

Aim of this section is to investigate whether it is possible to devise a suitable consensus algorithm guaranteeing that all the nodes of the network reach an agreement regarding the PDF of x . Here, by consensus algorithm we mean a distributed interaction rule specifying: (i) the information exchange between a node and its neighbors; (ii) the mechanism for updating the local PDF on the basis of the received information.

In order to formally state the average consensus problem of interest, the first important issue to be addressed is how to define the average PDF $\bar{p}(\cdot)$ given the agent PDFs $p^i(\cdot)$, $i \in \mathcal{N}$. To this end, it is convenient to recall that, given a set of points $\{x^1, \dots, x^N\}$ belonging to $\mathbb{R}^n$ and associated weights $\{\pi^1, \dots, \pi^N\}$ satisfying

$$\pi^i \geq 0, \quad \sum_{i=1}^N \pi^i = 1, \quad (1)$$

their weighted average $\bar{x} = \sum_{i=1}^N \pi^i x^i$ also satisfies the variational property of minimizing the weighted mean squared distance from such points, i.e.,

$$\bar{x} = \arg \min_{x \in \mathbb{R}^n} \sum_{i=1}^N \pi^i \|x - x^i\|^2 \quad (2)$$

where $\|\cdot\|$ denotes the Euclidean norm. As well known, there exist many ways of measuring the distance between two PDFs $p(\cdot)$ and $q(\cdot)$ in $\mathcal{P}$. From the information-theoretic point of view, the most typical choice corresponds to the *Kullback–Leibler divergence* (KLD) or *relative entropy* defined as

$$D_{\text{KL}}(p \parallel q) = \int p(x) \log \frac{p(x)}{q(x)} dx.$$

The KLD has many meaningful interpretations; for instance, in Bayesian statistics, it can be seen as the information gain achieved when moving from a prior PDF $q(\cdot)$ to a posterior PDF $p(\cdot)$. It is also worth noting that the KLD can be seen as the equivalent of the squared Euclidean distance in the space $\mathcal{P}$ of PDFs (in fact, they are both examples of Bregman divergences (Banerjee, Guo, & Wang, 2005)). In this connection, taking into account (2), it seems quite natural to introduce the following notion.

Definition 1. Given N PDFs $p^i(\cdot) \in \mathcal{P}$, and relative weights π^i satisfying (1), their weighted *Kullback–Leibler average* (KLA) is defined as follows

$$\bar{p} = \arg \inf_{p \in \mathcal{P}} \sum_{i=1}^N \pi^i D_{\text{KL}}(p \parallel p^i). \quad (3)$$

In particular, if $\pi^i = 1/N$ for $i = 1, \dots, N$, (3) provides the (unweighted) KLA which averages the agent PDFs giving to all of them the same degree of confidence. According to Definition 1, the average PDF is the one that minimizes the sum of the information gains from the initial PDFs. Thus, this choice is coherent with the *Principle of Minimum Discrimination Information* (PMDI) according to which the PDF which best represents the current state of knowledge is the one which produces an information gain as small as possible (the interested reader is referred to Akaike (1973) and Campbell (1970) for a discussion on such a principle and its relation with Gauss' principle and maximum likelihood estimation) or, in other words (Jaynes, 2003), “the probability assignment which most honestly describes what we know should be the most conservative assignment in the sense that it does not permit one to draw any conclusions not warranted by the data”. As it will be shown in the next sections, such a principle will lead to a quite remarkable property in the context of distributed state estimation for linear Gaussian systems.

Let now $p_\ell^i(\cdot) \in \mathcal{P}$ denote the PDF available at node i at the ℓ th iteration of the consensus algorithm, initialized from $p_0^i(\cdot) = p^i(\cdot)$. Then, thanks to Definition 1, the following average consensus problem can now be stated.

Problem 1. Find a consensus algorithm such that

$$\lim_{\ell \rightarrow \infty} p_\ell^i(x) = \bar{p}(x), \quad \forall i \in \mathcal{N},$$

where the asymptotic PDF $\bar{p}(\cdot)$ represents the KLA with uniform weights, i.e., with $\pi^i = 1/N$ for $i = 1, \dots, N$, of the initial PDFs $p^i(\cdot)$, $i \in \mathcal{N}$.

The following result concerning the KLA plays a key role in the solution of the above-stated problem.

Theorem 1. The weighted KLA defined in (3) turns out to be given by

$$\bar{p}(x) = \frac{\prod_{i=1}^N [p^i(x)]^{\pi^i}}{\int \prod_{i=1}^N [p^i(x)]^{\pi^i} dx}. \quad (4)$$

Proof. See the Appendix. ■

The above theorem states that the (weighted) KLA $\bar{p}(\cdot)$ is nothing but the normalized (weighted) geometric mean of the agent PDFs $p^i(\cdot)$; it is also called *Exponential Mixture Density (EMD)* in Uney, Clark, and Julier (2011). It is important to highlight that the fusion rule (4) actually coincides with the so called *Generalized Covariance Intersection (GCI)* which was introduced in Hurley (2002), Julier (2008), Julier, Bailey, and Uhlmann (2006), Mahler (2000) as a generalization of *Covariance Intersection* originally conceived (Julier & Uhlmann, 1997) for Gaussian PDFs. It is well known that GCI provides an information fusion that is robust with respect to the unknown correlations among the information sources.

To illustrate this point in a clear way, let us consider only two information sources Y^1 and Y^2 with associated conditional PDFs $p^i(x) = p(x|Y^i)$ for $i = 1, 2$. In general, the collective information set $Y^1 \cup Y^2$ can be decomposed into the union of three disjoint (independent) information sets as follows $Y^1 \cup Y^2 = (Y^1 \setminus Y^2) \cup (Y^2 \setminus Y^1) \cup (Y^1 \cap Y^2)$. Hence, the optimal fusion of $p^1(\cdot)$ and $p^2(\cdot)$ could be obtained as

$$\begin{aligned} p^{12}(x) &\propto p(x|Y^1 \cup Y^2) \\ &\propto p(x|Y^1 \setminus Y^2) p(x|Y^2 \setminus Y^1) p(x|Y^1 \cap Y^2) \\ &\propto \frac{p^1(x) p^2(x)}{p(x|Y^1 \cap Y^2)} \end{aligned} \quad (5)$$

if the PDF $p(x|Y^1 \cap Y^2)$ conditioned to the common information $Y^1 \cap Y^2$ were known (here the symbol $\propto$ stands for “proportional to” and the proportionality factor is determined by imposing that $p^{12}(\cdot)$ has unit integral over $\mathbb{R}^n$). However, in the considered framework wherein nodes repeatedly fuse information from their neighbors without any knowledge about the network topology, it is impossible to single out the common information and thus apply (5). Hence, some robust suboptimal fusion strategy has to be adopted. With this respect, the proposed KLA (or GCI) fusion provides

$$\begin{aligned} \bar{p}^{12}(x) &\triangleq \frac{[p^1(x)]^\pi [p^2(x)]^{1-\pi}}{\int [p^1(\xi)]^\pi [p^2(\xi)]^{1-\pi} d\xi} \\ &\propto [p(x|Y^1 \setminus Y^2)]^\pi [p(x|Y^2 \setminus Y^1)]^{1-\pi} p(x|Y^1 \cap Y^2) \end{aligned} \quad (6)$$

with $\pi \in [0, 1]$, where it can be seen that no double counting of common information occurs. The price to be paid is a conservative flattening $[p(Y^1 \setminus Y^2|x)]^\pi [p(Y^2 \setminus Y^1|x)]^{1-\pi}$ of independent information. Hence the fusion (4) under (1), turns out to be robust with respect to data incest.

In order to obtain the KLA (4) over the whole network in a distributed and scalable way, a consensus approach can be exploited. To this end, it is convenient to introduce two operators acting on PDFs.

More specifically, given two PDFs $p(\cdot)$ and $q(\cdot)$, their *addition* is defined as

$$p(x) \oplus q(x) = \frac{p(x) q(x)}{\int p(x) q(x) dx}.$$

Further, given a positive scalar π and a PDF $p(\cdot)$, the *scalar multiplication operator* is defined as

$$\pi \odot p(x) = \frac{[p(x)]^\pi}{\int [p(x)]^\pi dx}.$$

In view of such definitions, it can be checked that the two operators $\oplus$ and $\odot$ satisfy the following properties:

- (a) $(p \oplus q) \oplus r = p \oplus (q \oplus r) = p \oplus q \oplus r$
- (b) $p \oplus q = q \oplus p$
- (c) $(\alpha \beta) \odot p = \alpha \odot (\beta \odot p)$
- (d) $1 \odot p = p$
- (e) $\alpha \odot (p \oplus q) = (\alpha \odot p) \oplus (\alpha \odot q)$
- (f) $(\alpha + \beta) \odot p = (\alpha \odot p) \oplus (\beta \odot p)$

where p, q, r are PDFs and α, β are positive scalars. Further, the identity (4) with $\pi^i = \frac{1}{N}$ for all i can be rewritten as

$$\bar{p}(x) = \bigoplus_{i \in \mathcal{N}} \left(\frac{1}{N} \odot p^i(x) \right). \quad (7)$$

Then, the collective KLA (7), which would require all the local agent PDFs to be available, can be computed in a distributed and scalable way by applying one of the many existing consensus algorithms for distributed averaging. In this connection, the simplest distributed averaging algorithms consist of updating the local data via convex combination with the data received by the neighbors (Olfati-Saber et al., 2007; Xiao et al., 2005). In this case, such a choice gives rise to the following consensus algorithm:

$$p_{\ell+1}^i(x) = \bigoplus_{j \in \mathcal{N}^i} \left(\pi^{ij} \odot p_{\ell}^j(x) \right), \quad \forall i \in \mathcal{N} \quad (8)$$

where $p_0^i(x) = p^i(x)$ and π^{ij} are suitable non-negative weights such that

$$\sum_{j \in \mathcal{N}^i} \pi^{ij} = 1, \quad \forall i \in \mathcal{N}.$$

In fact, thanks to properties (a)–(f), it can be seen that²

$$p_{\ell}^i(x) = \bigoplus_{j \in \mathcal{N}} \left(\pi_{\ell}^{ij} \odot p^j(x) \right), \quad \forall i \in \mathcal{N} \quad (9)$$

where π_{ℓ}^{ij} is defined as the element (i, j) of the matrix Π^{ℓ} and Π is the consensus matrix whose generic (i, j) -element coincides with the consensus weight π^{ij} (if $j \notin \mathcal{N}^i$ then π^{ij} is taken as 0). With this respect, recall that when the consensus weights π^{ij} are chosen so as to ensure that the matrix Π is primitive and doubly stochastic,³ one has

$$\lim_{\ell \rightarrow +\infty} \pi_{\ell}^{ij} = \frac{1}{N}, \quad \forall i, j \in \mathcal{N}.$$

Hence, as the number of consensus steps increases, each local PDF “tends” to the collective KLA (7).

As well known (Calafiore & Abrate, 2009), a necessary condition for the matrix Π to be primitive is that the graph $\mathcal{G}$ associated with the sensor network be strongly connected, i.e. for any pair of nodes $i, j \in \mathcal{N}$ there must be a path from i to j and vice versa. Further,

² Notice that the scalar multiplication operator is defined only for strictly positive scalars. However, in Eq. (9) it is admitted that some of the scalar weights π_{ℓ}^{ij} be zero. The understanding is that, whenever π_{ℓ}^{ij} is equal to zero, the corresponding PDF $p^j(x)$ is omitted from the addition. This can always be done, since for each $i \in \mathcal{N}$ and each ℓ , at least one of the weights π_{ℓ}^{ij} is strictly positive.

³ Recall that a non-negative square matrix Π is doubly stochastic if all its rows and columns sum up to 1. Further, it is primitive if there exists an integer m such that all the elements of Π^m are strictly positive.

when the graph $\mathcal{G}$ is undirected, a possible choice ensuring that the consensus weight matrix be doubly stochastic is given by the so-called *Metropolis weights* (Calafiore & Abrate, 2009; Xiao et al., 2005)

$$\pi^{ij} = \frac{1}{\max\{|\mathcal{N}^i|, |\mathcal{N}^j|\}}, \quad i \in \mathcal{N}, j \in \mathcal{N}^i, i \neq j$$

$$\pi^{ii} = 1 - \sum_{j \in \mathcal{N}^i, j \neq i} \pi^{ij}.$$

Notice that the consensus on generic PDFs introduced in this section provides a consistent, distributed and scalable way to fuse information from multiple interconnected local estimators and has general applicability to nonlinear and/or non-Gaussian distributed estimation problems. Further, a distinctive feature of the proposed approach, as compared to usual applications of GCI, is that here the regional fusions are performed with the objective of achieving a collective goal (i.e., computation of the collective KLA).

Clearly, the convergence rate of the consensus algorithm (8) to the KLA depends on the choice of the weight matrix Π , and in particular on the second largest eigenvalue of such a matrix (Xiao & Boyd, 2004). Bounds on the worst-case convergence time for various choices of the consensus weights (including the Metropolis weights) can be found in Olshevsky and Tsitsiklis (2009). Further, when some information on the network topology is a-priori available, the optimal consensus weights achieving the fastest convergence rate can be determined by solving a convex optimization problem (Xiao & Boyd, 2004). However, how to determine, or at least approximate, such optimal weights in a completely distributed fashion is still an open problem.

The Gaussian case

In the remainder of this section, for the sake of brevity, the attention will be restricted to a particular parametric family of PDFs, namely, normal distributions. However, similar considerations could be made also for all those parametric families for which the KLD can be computed in closed form (such families include those belonging to the exponential class of density functions like the exponential distribution, the binomial distribution, etc.).

Let now all the local PDFs $p^i(\cdot)$ take the form

$$p^i(x) = \mathcal{N}(x; \mu^i, \Sigma^i)$$

$$\triangleq \frac{1}{\sqrt{\det(2\pi \Sigma^i)}} e^{-\frac{1}{2}(x-\mu^i)^\top (\Sigma^i)^{-1} (x-\mu^i)} \quad (10)$$

where $\mu^i \in \mathbb{R}^n$ is the mean and $\Sigma^i \in \mathbb{R}^{n \times n}$ is the (positive definite) covariance matrix. Then, the following result can be stated.

Theorem 2. Given Gaussian PDFs as in (10) and weights π^i satisfying (1), their weighted KLA in (4) takes the form $\bar{p}(\cdot) = \mathcal{N}(\cdot; \bar{\mu}, \bar{\Sigma})$ where the mean $\bar{\mu}$ and the covariance matrix $\bar{\Sigma}$ can be computed from the following algebraic relationships

$$\bar{\Sigma}^{-1} = \sum_{i=1}^N \pi^i (\Sigma^i)^{-1}, \quad (11)$$

$$\bar{\Sigma}^{-1} \bar{\mu} = \sum_{i=1}^N \pi^i (\Sigma^i)^{-1} \mu^i. \quad (12)$$

Proof. See the Appendix. ■

Recalling that $\Omega^i \triangleq (\Sigma^i)^{-1}$ and $q^i \triangleq (\Sigma^i)^{-1} \mu^i$ are the *information matrix* and *information vector*, respectively, associated with the Gaussian PDF (10), Theorem 2 states that the KLA of N Gaussian

PDFs can be simply obtained by averaging their information matrices and information vectors. An important consequence of this state of affairs is that Problem 1 can be readily solved by iterating regional weighted averages through the consensus algorithm⁴

$$\Omega_{\ell+1}^i = \sum_{j \in \mathcal{N}^i} \pi^{ij} \Omega_{\ell}^j, \quad i \in \mathcal{N}, \ell = 0, 1, \dots \quad (13)$$

$$q_{\ell+1}^i = \sum_{j \in \mathcal{N}^i} \pi^{ij} q_{\ell}^j, \quad i \in \mathcal{N}, \ell = 0, 1, \dots \quad (14)$$

Clearly, the recursion is initialized by letting

$$\Omega_0^i = \Omega^i, \quad i \in \mathcal{N} \quad (15)$$

$$q_0^i = q^i, \quad i \in \mathcal{N}. \quad (16)$$

Recalling the previous definition of consensus matrix Π , the following proposition derives from well-known results on distributed averaging (Olfati-Saber et al., 2007; Xiao et al., 2005).

Proposition 1. Let the consensus matrix Π be primitive and doubly stochastic. Then, the consensus algorithm (13)–(16) asymptotically yields the KLA of the initial local PDFs in that

$$\lim_{\ell \rightarrow \infty} \Omega_{\ell}^i = \bar{\Omega} \triangleq \frac{1}{N} \sum_{i=1}^N (\Sigma^i)^{-1}$$

$$\lim_{\ell \rightarrow \infty} q_{\ell}^i = \bar{q} \triangleq \frac{1}{N} \sum_{i=1}^N (\Sigma^i)^{-1} \mu^i.$$

3. Application to distributed state estimation

In this section, the developments of the previous section are applied to distributed state estimation. To this end, let us consider a discrete-time dynamical system

$$x_{k+1} = f(x_k) + w_k$$

where k is the discrete time index, $x_k \in \mathbb{R}^n$ is the system state and $w_k \in \mathbb{R}^n$ is the process noise. The initial state x_0 of the system is unknown but distributed according to a known PDF $p_{0|-1}(\cdot)$. It is supposed that the sequence $\{w_k\}$ is generated by a zero-mean white stochastic process with known PDF $p_w(\cdot)$. Further, suppose that, at each time $k = 0, 1, \dots$ each node $i \in \mathcal{N}$ of the network collects a measurement

$$y_k^i = h^i(x_k) + v_k^i$$

of the state vector x_k , where $v_k^i \in \mathbb{R}^{n_y}$ is generated by a zero-mean white stochastic process with known PDF $p_{v^i}(\cdot)$ (the initial state x_0 and the sequences $\{w_k\}$ and $\{v_k^i\}$ are supposed to be mutually independent).

If no communication between the nodes of the network were possible, the solution of the local state estimation problem would yield the well-known Bayesian filtering recursion:

$$p_{0|-1}^i(x) = p_{0|-1}(x), \quad (17)$$

$$p_{k|k}^i(x) = \frac{p_{v^i}[y_k^i - h^i(x)] p_{k|k-1}^i(x)}{\int p_{v^i}[y_k^i - h^i(\xi)] p_{k|k-1}^i(\xi) d\xi}, \quad (18)$$

$$p_{k+1|k}^i(x) = \int p_w[x - f(\xi)] p_{k|k}^i(\xi) d\xi, \quad (19)$$

⁴ Along the lines of Theorem 2 it could be seen that the update rules (13) and (14) correspond to computing a weighted KLA between the local PDF and the PDFs of the neighbors.

for $k = 0, 1, \dots$, where $p_{k|t}^i(\cdot)$ represents the PDF of x_k conditioned to all the measurements collected at node i up to time t .

If, however, a communication structure is available as described in the previous section, then each node can improve its local estimate by fusing the local information with the one received from its neighbors. More specifically, one can perform at each time instant a certain number, say L , of consensus steps on the posterior PDFs $p_{k|k}^i(\cdot)$, $i \in \mathcal{N}$, in order to compute in a distributed fashion their KLA. This idea leads to the following Distributed State Estimation Algorithm with Consensus on the Posteriors (DSEA-CP).

Algorithm 1. At each time $k = 0, 1, \dots$, for each node $i \in \mathcal{N}$:

- (1) collect the local measurement y_k^i and update the local PDF $p_{k|k-1}^i(\cdot)$ via Eq. (18) to obtain the local posterior $p_{k|k,0}^i$;
- (2) perform L steps of consensus on the KLA average to obtain the fused posterior $p_{k|k,L}^i$;
- (3) compute the local prior $p_{k+1|k}^i$ from $p_{k|k}^i \triangleq p_{k|k,L}^i$ via Eq. (19).

In view of the discussion of the previous section, the rationale for combining the local posteriors according to the KLA paradigm is to counteract the deleterious effects of data incest by avoiding any double-counting of information. A formal analysis of this property will be provided in Section 4 for the linear-Gaussian case.

3.1. The linear-Gaussian case: covariance intersection revisited

Suppose now that both the system dynamics and the measurement equations are linear, i.e., consider a discrete-time linear system

$$x_{k+1} = Ax_k + w_k \quad (20)$$

whose state is measured by N linear sensors

$$y_k^i = C^i x_k + v_k^i, \quad i = 1, \dots, N. \quad (21)$$

Further, let the initial state, the process disturbance and all the measurement noises be normally distributed, i.e.,

$$\begin{aligned} p_0(x) &= \mathcal{N}(x; \hat{x}_{0|-1}, P_{0|-1}), \\ p_w(w) &= \mathcal{N}(w; 0, Q), \\ p_{v^i}(v^i) &= \mathcal{N}(v^i; 0, R^i), \quad i = 1, \dots, N, \end{aligned}$$

where $\hat{x}_{0|-1}$ is a known vector and $P_{0|-1}, Q, R^i$, $i = 1, \dots, N$ are known positive definite matrices. As well known, in this case, the Bayesian filtering recursion (17)–(19) admits a closed-form solution in that, for each node i , one has

$$\begin{aligned} p_{k|k}^i(x) &= \mathcal{N}(x; \hat{x}_{k|k}^i, P_{k|k}^i), \\ p_{k+1|k}^i(x) &= \mathcal{N}(x; \hat{x}_{k+1|k}^i, P_{k+1|k}^i) \end{aligned}$$

where the mean vectors $\hat{x}_{k|k}^i$ and $\hat{x}_{k+1|k}^i$ and the covariance matrices $P_{k|k}^i$ and $P_{k+1|k}^i$ can be recursively computed by means of the Kalman filter recursion. In view of the developments of the previous section, in order to streamline the presentation as well as the stability analysis, it is convenient to consider the *information form* of the Kalman filter recursion whereby, instead of the mean vectors and the covariance matrices, the information matrices

$$\Omega_{k|k}^i = (P_{k|k}^i)^{-1}, \quad \Omega_{k+1|k}^i = (P_{k+1|k}^i)^{-1}$$

and information vectors

$$q_{k|k}^i = \Omega_{k|k}^i \hat{x}_{k|k}^i, \quad q_{k+1|k}^i = \Omega_{k+1|k}^i \hat{x}_{k+1|k}^i$$

are recursively propagated. With this choice, the update step can be simply written as

$$q_{k|k}^i = q_{k|k-1}^i + (C^i)^\top (R^i)^{-1} y_k^i, \quad (22)$$

$$\Omega_{k|k}^i = \Omega_{k|k-1}^i + (C^i)^\top (R^i)^{-1} C^i, \quad (23)$$

whereas the prediction step takes the form

$$q_{k+1|k}^i = A^{-\top} \left[I - \Omega_{k|k}^i (\Omega_{k|k}^i + A^\top Q^{-1} A)^{-1} \right] q_{k|k}^i, \quad (24)$$

$$\begin{aligned} \Omega_{k+1|k}^i &= A^{-\top} \Omega_{k|k}^i A^{-1} \\ &\quad - A^{-\top} \Omega_{k|k}^i (\Omega_{k|k}^i + A^\top Q^{-1} A)^{-1} \Omega_{k|k}^i A^{-1}. \end{aligned} \quad (25)$$

Notice that, for Eqs. (24) and (25) to make sense, the system matrix A has to be invertible. This hypothesis is automatically satisfied in sampled-data systems wherein the matrix A is obtained by discretization of a continuous-time system matrix. Further, in the other cases, such a limitation could be easily overcome by considering different information filter algorithms (e.g. the square-root information filter proposed by Chisci and Zappa (1992)). However, these generalizations are not addressed here in order to keep the presentation as streamlined as possible.

By exploiting the above-defined information filter as well as the consensus algorithm (13)–(14), it is possible to specialize Algorithm 1 as follows.

Algorithm 2. At each time $k = 0, 1, \dots$, for each node $i \in \mathcal{N}$:

- (1) collect the local measurement y_k^i and update the local information pair $(q_{k|k-1}^i, \Omega_{k|k-1}^i)$ via Eqs. (22)–(23) to obtain the local posterior information pair $(q_{k|k,0}^i, \Omega_{k|k,0}^i)$;
- (2) perform L steps of consensus on the KLA average

$$\Omega_{k|k,\ell+1}^i = \sum_{j \in \mathcal{N}^i} \pi^{i,j} \Omega_{k|k,\ell}^j, \quad \ell = 0, \dots, L-1 \quad (26)$$

$$q_{k|k,\ell+1}^i = \sum_{j \in \mathcal{N}^i} \pi^{i,j} q_{k|k,\ell}^j, \quad \ell = 0, \dots, L-1 \quad (27)$$

to obtain the fused information pair $(q_{k|k}^i, \Omega_{k|k}^i) \triangleq (q_{k|k,L}^i, \Omega_{k|k,L}^i)$;

- (3) compute the local prior information pair $(q_{k+1|k}^i, \Omega_{k+1|k}^i)$ from $(q_{k|k,L}^i, \Omega_{k|k,L}^i)$ via Eqs. (24)–(25).

In principle, the algorithm should be initialized by letting

$$\Omega_{0|-1}^i = P_{0|-1}^{-1}, \quad i \in \mathcal{N}$$

$$q_{0|-1}^i = P_{0|-1}^{-1} \hat{x}_{0|-1}, \quad i \in \mathcal{N}.$$

However, since it is quite unrealistic to assume that all the nodes of the network share the same a priori information on the initial state x_0 , a more practical initialization can be obtained by letting

$$\Omega_{0|-1}^i = 0, \quad i \in \mathcal{N} \quad (28)$$

$$q_{0|-1}^i = 0, \quad i \in \mathcal{N}, \quad (29)$$

which would amount to assuming that no a priori information is available.

It is important to point out that the idea of fusing the information received from multiple sensors by making a convex combination of the information matrices and vectors as in (26)–(27) is not novel in the information fusion literature (albeit the derivation proposed here is quite different from the existing ones). In fact, such a fusion rule is known in the literature as *Covariance Intersection* and dates back to over a decade ago (Chen et al., 2002; Julier & Uhlmann, 1997). In this connection, a contribution of this paper consists of showing that such a widely-used fusion rule naturally arises as a step of a consensus algorithm for computing, in a distributed fashion, the KLA of the local posterior PDFs.

Notice also that the proposed algorithm for distributed state estimation presents some important differences when compared

with other consensus-based approaches presented in the literature. In fact, the existing approaches can be mainly divided into two families: in the first group (Cattivelli & Sayed, 2010; Olfati-Saber, 2009; Stankovic et al., 2009), a consensus among the network nodes is sought by averaging out only the state estimates, thus disregarding the covariance matrices as well as the form of the local PDFs in the nonlinear or non-Gaussian case; the approaches belonging to the second group (Kamgarpour & Tomlin, 2008; Olfati-Saber, 2007) adopt a different point of view and perform a consensus on the local innovation pairs $((C^i)^T (R^i)^{-1} y_k^i, (C^i)^T (R^i)^{-1} C^i)$ so as to approximate, in a distributed way, the correction step of the centralized Kalman filter. The main limitation of the approaches of this second group is that stability can be guaranteed only when a sufficiently large number of consensus steps is performed at each time step, so that the local information provided by the innovation pairs has time to spread throughout the whole network (see also the numerical example of Section 5). Further, this kind of approach relies on the assumption that the measurement errors relative to different sensors are mutually independent and can only be applied to Kalman-like filters. As will be also clarified in the following section, the proposed approach does not suffer from such limitations and enjoys strong stability properties thanks to the fact that the whole posterior PDFs are combined (rather than just the state estimates or the innovation pairs) according to the KLA paradigm. On the other hand, the proposed algorithm cannot be expected to approach the centralized Kalman-filter since the adopted fusion rule follows a cautious strategy so as to achieve robustness with respect to data incest.

4. Stability analysis

In this section, the stability properties of the proposed distributed state estimation algorithm are analyzed. To this end, in order to ensure the well-posedness of the state estimation problem, the following preliminary assumptions are needed.

A1 The system matrix A is invertible.

A2 The system is collectively observable, i.e. the pair (A, C) is observable where $C \triangleq \text{col}(C^1, \dots, C^N)$.

Some comments on assumption A2 are in order. The reason for requiring collective observability rather than collective detectability serves the purpose of simplifying the stability analysis. However, this can be done without loss of generality, since by means of a suitable linear change of variables

$$x_k = Tz_k$$

one can always consider an observability decomposition of the state space-dynamics of the form

$$z_{k+1} = \tilde{A}z_k + \tilde{w}_k,$$

$$y_k^i = \tilde{C}^i z_k + v_k^i, \quad i \in \mathcal{N}$$

with

$$z_k = \begin{bmatrix} z_k^o \\ z_k^{uo} \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} A^o & 0 \\ A^m & A^{uo} \end{bmatrix}, \quad \tilde{C}^i = [C^{i,o} \ 0],$$

and the pair (A^o, C^o) , $C^o \triangleq \text{col}(C^{1,o}, \dots, C^{N,o})$, observable. Afterwards, one can apply the proposed distributed estimation algorithm only to the collectively observable subsystem in order to estimate z_k^o . Then, an estimate of the unobservable state z_k^{uo} can be recursively obtained as

$$\hat{z}_{k|k}^{uo} = A^{uo} \hat{z}_{k-1|k-1}^{uo} + A^m \hat{z}_{k-1|k-1}^o.$$

Since, under collective detectability, the matrix A^{uo} is Schur stable, mean-square stability of $\hat{z}_{k|k}^o - z_k^o$ automatically implies also mean-square stability of $\hat{z}_{k|k}^{uo} - z_k^{uo}$.

One of the most basic features that a recursive state estimation algorithm should enjoy is that the true estimation errors be consistent with their predicted statistics. In fact, as well known (Jazwinski, 1970), divergence of the estimation error can take place when the covariance matrix becomes too small or optimistic (because in this case subsequent observations tend to be ignored). In this connection, the following definition can be introduced (Jazwinski, 1970; Julier & Uhlmann, 1997).

Definition 2. Consider a random vector x . Further, let $\hat{x}$ be an unbiased estimate of x and P an estimate of the corresponding error covariance. Then, the pair $(\hat{x}, P)$ is said to be *consistent* if

$$\mathbb{E} \{ (x - \hat{x})(x - \hat{x})^T \} \leq P. \quad (30)$$

In words, according to inequality (30), consistency amounts to requiring that the estimated error covariance P be an upper bound (in the positive definite sense) of the true error covariance. This property becomes even more important in distributed state estimation because the unaware reuse of the same data due to the presence of loops within the network as well as the possible correlation between measurements of different sensors can lead to inconsistency and divergence. In fact, this was the primary motivation that led to the development of the Covariance Intersection fusion rule.

Notice now that, if one considers the information pair $(q, \Omega) = (P^{-1}\hat{x}, P^{-1})$, inequality (30) can be rewritten as

$$\Omega \leq (\mathbb{E} \{ (x - \Omega^{-1}q)(x - \Omega^{-1}q)^T \})^{-1}.$$

Then, taking into account the relationship between Algorithm 2 and Covariance Intersection, the following result can be stated.

Theorem 3. Let assumption A1 hold and let the distributed state estimation Algorithm 2 be adopted with the initialization (28)–(29). Then, for each $k = 0, 1, \dots$ and each $i \in \mathcal{N}$, the information pair $(q_{k|k}^i, \Omega_{k|k}^i)$ is consistent in that

$$\Omega_{k|k}^i \leq (\mathbb{E} \{ (x_k - \hat{x}_{k|k}^i)(x_k - \hat{x}_{k|k}^i)^T \})^{-1} \quad (31)$$

with $\hat{x}_{k|k}^i \triangleq (\Omega_{k|k}^i)^+ q_{k|k}^i$.⁵

Proof. See the Appendix. ■

Theorem 3 points out that, in the linear case, thanks to the adherence to the PMDI, the proposed distributed state estimation algorithm avoids double-counting and preserves consistency of all the information pairs. Further, in view of Theorem 3, in order to prove the boundedness of the error covariance $\mathbb{E} \{ (x_k - \hat{x}_{k|k}^i)(x_k - \hat{x}_{k|k}^i)^T \}$ it is sufficient to show that asymptotically the information matrix $\Omega_{k|k}^i$ is bounded below by some positive definite matrix (or, equivalently, to show that $P_{k|k}^i = (\Omega_{k|k}^i)^{-1}$ is asymptotically bounded above by some constant matrix). In this connection, the following theorem, representing the main result of this section, can be stated.

Theorem 4. Let assumption A1–A2 hold and let the distributed state estimation Algorithm 2 be adopted with the initialization (28)–(29). Then, if the consensus matrix Π is primitive and the number L of consensus iterations is greater than 0, there exist a time instant $\bar{k}$ and a positive definite matrix $\tilde{\Omega}$ such that

$$0 < \tilde{\Omega} \leq \Omega_{k|k}^i, \quad \forall i \in \mathcal{N} \text{ and } \forall k \geq \bar{k}.$$

⁵ Here, the Moore–Penrose pseudoinverse of the information matrix $\Omega_{k|k}^i$, denoted as $(\Omega_{k|k}^i)^+$, is considered to account for the fact that, under the uninformative prior (28)–(29), such a matrix is singular during the first time instants.

As a consequence, the estimation error is asymptotically bounded in mean square in that

$$\limsup_{k \rightarrow \infty} \mathbb{E} \{ (x_k - \hat{x}_{k|k}^i)^\top (x_k - \hat{x}_{k|k}^i) \} \leq \text{tr} \{ \tilde{\Sigma}^{-1} \}.$$

Proof. See the Appendix. ■

A few remarks on Theorem 4 are in order. First of all, it is important to point out that such a stability result relies only on the assumption of *collective* observability, i.e. observability from the whole network (which can be further relaxed to collective detectability as discussed at the beginning of this section). This has to be contrasted with those stability analyses which require some sort of *local* observability or detectability condition (see, for instance, Cattivelli & Sayed, 2010; Olfati-Saber, 2009; Stankovic et al., 2009), thus posing strong assumptions on the quality of the sensors and limiting the practical applicability. With this respect, it is worth noting that the findings of this paper are consistent with the recent independent results of Ugrinovskii (2013), where it is shown that collective detectability is indeed a necessary and sufficient condition for the existence of distributed observers of the form of Ugrinovskii (2011) with a stable error dynamics.

It is also worth noting that such a stability result holds regardless of the possible correlation between the measurements coming from different sensors (which is supposed to be unknown). Further, only one consensus step per iteration is sufficient for stability.

Another important feature of the proposed distributed state estimation algorithm is that, when the system dynamics as well as all the measurement equations are noise-free, it is possible to show that all the local estimation errors $x_k - \hat{x}_{k|k}^i$ converge asymptotically to zero. To see this, it is convenient to consider the time-varying quadratic Lyapunov functions

$$\mathcal{L}_k^i(x) \triangleq x^\top \Omega_{k|k-1}^i x, \quad i \in \mathcal{N}$$

and to define the vector

$$\mathcal{L}_k = \text{col} (\mathcal{L}_k^1(x_k - \hat{x}_{k|k-1}^1), \dots, \mathcal{L}_k^N(x_k - \hat{x}_{k|k-1}^N)).$$

Then, the following theorem can be stated.

Theorem 5. Let the system dynamics (20) and the measurement equations (21) be noise-free, i.e.,

$$w_k = 0, \quad v_k^i = 0, \quad i \in \mathcal{N}$$

for $k = 0, 1, \dots$. Then, under the same assumptions of Theorem 4, there exist a time instant $\bar{k}$ and a positive real $\beta < 1$ such that

$$\mathcal{L}_{k+1} \leq \beta \Pi^L \mathcal{L}_k,$$

for $k = \bar{k}, \bar{k} + 1, \dots$

Proof. See the Appendix. ■

Since, by construction, the consensus matrix Π is row stochastic, its spectral radius is always 1. As a consequence, for $\beta < 1$, the matrix $\beta \Pi^L$ turns out to be Schur stable. This, in turn, implies that all the components of the vector $\mathcal{L}_k$ vanish as k goes to infinity. Hence, given that all the matrices $\Omega_{k|k-1}^i$ are asymptotically bounded from below by some positive definite matrix (this is a straightforward consequence of Theorem 4), the following corollary to Theorem 5 can be readily stated.

Corollary 1. Under the same assumptions of Theorem 5, the proposed distributed state estimation algorithm yields an asymptotic observer in each node of the network, in that

$$\lim_{k \rightarrow \infty} x_k - \hat{x}_{k|k}^i = 0, \quad \forall i \in \mathcal{N}.$$

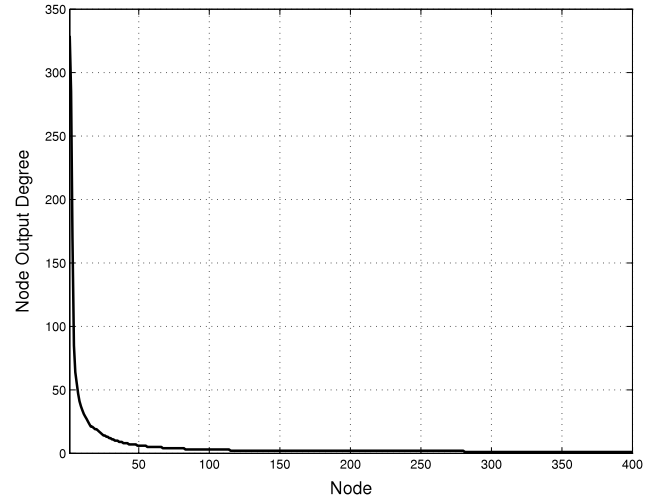


Fig. 1. Out-degree of each node of the network.

5. Simulation results

The goal of this section is to evaluate the performance of the proposed distributed state estimation algorithm. To this end, let us consider an object whose motion is described by the kinematic nearly-constant velocity model:

$$x_{k+1} = \begin{bmatrix} 1 & \Delta & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta \\ 0 & 0 & 0 & 1 \end{bmatrix} x_k + w_k$$

where $\Delta = 4$ is the sampling interval and the unknown state vector is given by the position and velocity components along the coordinate axes, i.e., $x = [p_x, v_x, p_y, v_y]'$. According to Bar-Shalom, Li, and Kirubarajan (2001, p. 270, Eq. 6.2.2–12), the covariance matrix of process noise has been assumed equal to $Q = G q^2$ with

$$G = \begin{bmatrix} \Delta^3/3 & \Delta^2/2 & 0 & 0 \\ \Delta^2/2 & \Delta & 0 & 0 \\ 0 & 0 & \Delta^3/3 & \Delta^2/2 \\ 0 & 0 & \Delta^2/2 & \Delta \end{bmatrix}$$

and $q^2 = 5$.

Scenario 1

In the first simulation scenario, the kinematic state of the object is estimated by means of a multi-sensor network with power-law connectivity (i.e., the degree distribution follows a power law as shown in Fig. 1). The network consists of a total of 400 nodes, 10 of which are equipped with a position sensor whereas the remaining 390 only possess information processing and communication capabilities. The sensor nodes have been chosen among those with only one link. The considered scenario is depicted in Fig. 2.

Each of the 10 sensor nodes provides measurements of the object's position in Cartesian coordinates, i.e.,

$$y_k^i = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} x_k + v_k^i.$$

For the remaining nodes, the measurement equation can be thought of as $y_k^i = v_k^i$. The covariance matrices of measurement noise have been assumed equal to $R^i = \text{diag}(r^2, r^2)$ where $r = 15$. All the measurements are mutually uncorrelated.

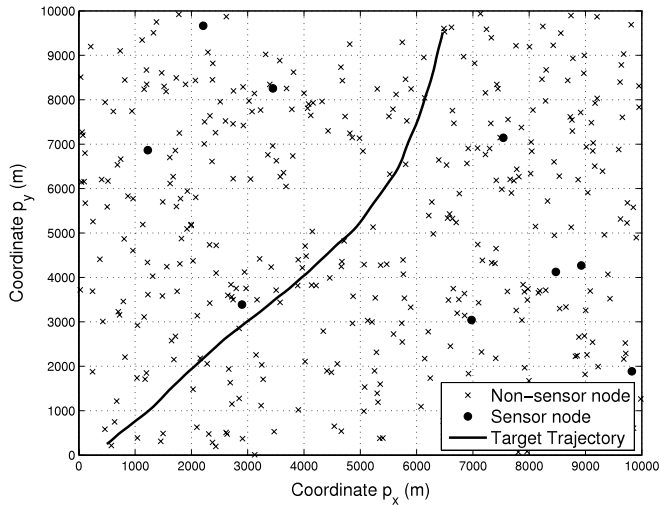


Fig. 2. Graphical representation of the first simulation scenario. In order to keep the figure easy to read, the network links are not depicted here.

The performance of the DSEA-CP described in Section 3.1 has been analyzed for different values of the parameter L (representing the number of consensus steps per iteration) and compared to that of the Distributed State Estimation Algorithm with Consensus on the Measurements (DSEA-CM), i.e. algorithm 1 of Olfati-Saber (2007), and to that of the Centralized Kalman filter (CKF), obtained by considering all the measurements available to each node. The comparison between the algorithms has been carried out via Monte Carlo simulations with 200 independent runs obtained for randomly generated trajectories by varying the measurement and process noise realizations.

Fig. 3 shows the behavior of the Root Mean Square Error (RMSE), averaged over all the nodes, of the considered filters. It can be seen that the performance of the DSEA-CM depends crucially on the number of consensus steps and, in particular, it needs a minimum number of consensus steps per iteration in order to avoid divergence (in this case, since the network is highly connected, two steps are required). On the other hand, despite the lack of local observability, the proposed DSEA-CP algorithm provides reasonable performance even for $L = 1$. Further, as L increases, it performs closer to the CKF (that exploits collective information). It is believed that such satisfactory performance is made possible by the fact that the whole posterior PDF is considered for distributed averaging rather than just the estimates or the innovations, as in most of the previously proposed consensus-based distributed estimation algorithms. It is worth to point out that the improvement of DSEA-CP over DSEA-CM comes at no extra cost as both involve consensus on an $n \times n$ matrix and an $n \times 1$ vector, thus having comparable computational burden.

Scenario 2

In the second simulation scenario, a different multi-sensor network is considered. The network consists of a total of 35 nodes, 5 of which are equipped with a nonlinear position sensor whereas the remaining 30 only possess information processing and communication capabilities. The position sensors, from now on indicated by the acronym TOA (Time Of Arrival), measure the relative distance between the sensor and the moving object and are characterized by the nonlinear measurement equation

$$y_k^i = \sqrt{(p_{x,k} - p_x^i)^2 + (p_{y,k} - p_y^i)^2} + v_k^i$$

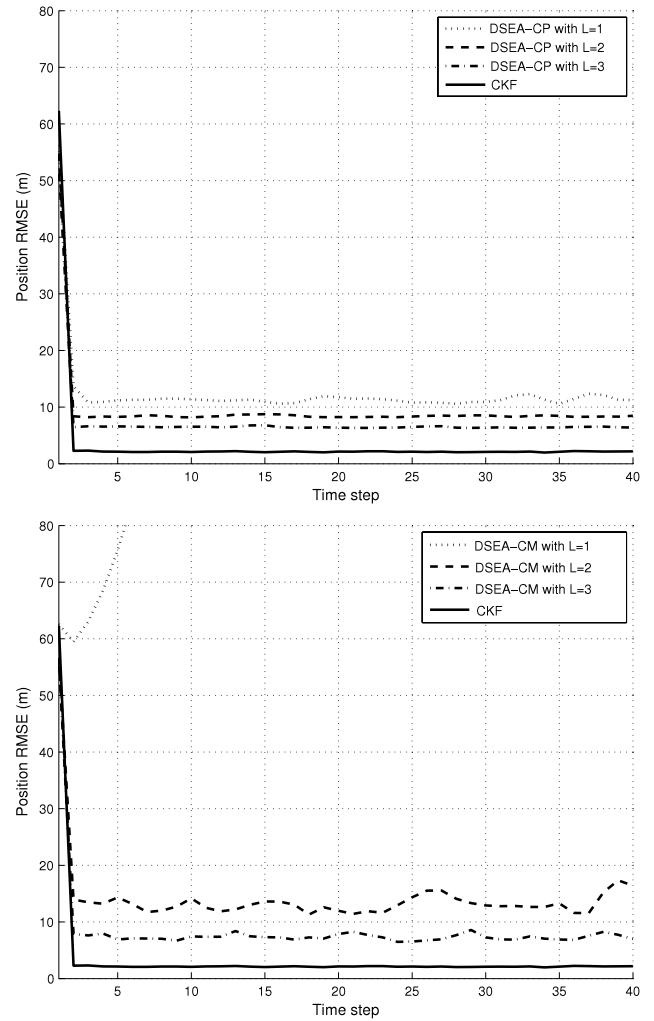


Fig. 3. Position RMSE of the considered filters in the first simulation scenario.

where (p_x^i, p_y^i) denotes the position of the i th sensor. The variance of the measurement noise has been assumed equal to 50^2 . The network topology is geometric and, specifically, two nodes are connected if and only if their distance does not exceed the connectivity radius $r_{\max} = 1493$. A graphical representation of this second scenario is provided in Fig. 4. Notice that, in this case, a single sensor does not provide observability of the whole system state and, hence, cooperation is needed in order to reliably estimate the kinematic state of the target.

The DSEA-CP has been implemented by approximating the true conditional PDFs with Gaussians. To this end, the updated mean and covariance in the local correction step have been computed by resorting to the Unscented Kalman filter (UKF) (Julier & Uhlmann, 2004). Performance of the DSEA-CP has been evaluated over 200 independent Monte Carlo runs for different values of L and compared to that of the centralized UKF (CUKF). The behaviors of the Root Mean Square Error (RMSE) of the considered filters, averaged over all the nodes, are shown in Fig. 5. As it can be seen, also in this nonlinear setting, the proposed distributed state estimation algorithm provides a bounded estimation error even when only one consensus step per iteration ($L = 1$) is performed. Further, performance improves as L increases.

It is worth to remind that, although in the considered scenario the UKF-based DSEA-CP seems to work well with a nonlinear observation model, the performance is likely to be highly dependent on how well the UKF approximates the target distribution. Further,

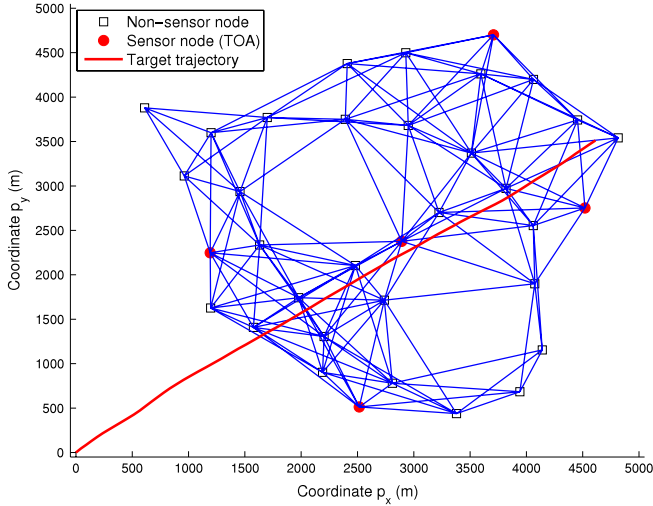


Fig. 4. Graphical representation of the second simulation scenario.

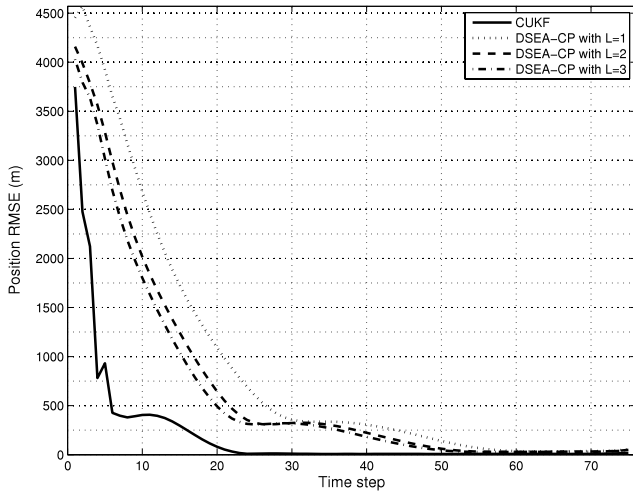


Fig. 5. Position RMSE of the considered filters in the second simulation scenario.

the stability results of Section 4 do not directly carry over to the nonlinear setting.

6. Conclusions

The main novelty of this paper has been to introduce consensus on probability density functions (PDFs) so as to compute in a distributed and scalable fashion their Kullback–Leibler average. This has allowed to derive a novel consensus-based networked Kalman filter that, in a linear setting, guarantees filter stability in all network nodes even for a single consensus step. It is worth to point out that the obtained stability result relies only on network connectivity and collective detectability, i.e. system detectability from the observations of the whole sensor network.

While in principle the proposed approach can deal with PDFs of arbitrary form, a closed-form expression for the consensus algorithm exists only in special cases (for instance, when all the local PDFs belong to the same exponential family). Hence, the treatment of a nonlinear non-Gaussian setting requires in general some sort of approximation, whose accuracy crucially affects performance and stability. This issue will be the subject of future research. Future work will also concern networked state estimation of spatially distributed systems.

Appendix. Proofs

Proof of Theorem 1. Let

$$\tilde{p}(x) = \prod_i [p^i(x)]^{\pi^i}, \quad c = \int \tilde{p}(x) dx$$

so that the PDF in (4) can be expressed as $\bar{p}(x) = \tilde{p}(x)/c$. Then, the cost to be minimized by the KLA is

$$\begin{aligned} J(p) &\triangleq \sum_{i=1}^N \pi^i D_{KL}(p \parallel p^i) \\ &= \sum_{i=1}^N \pi^i \int p(x) \log \frac{p(x)}{p^i(x)} dx \\ &= \int p(x) \left[\sum_{i=1}^N \pi^i \log \frac{p(x)}{p^i(x)} \right] dx \\ &= \int p(x) \log \frac{p(x)}{\prod_{i=1}^N p^i(x)^{\pi^i}} dx \\ &= \int p(x) \log \frac{p(x)}{\tilde{p}(x)} dx \\ &= \int p(x) \log \frac{p(x)}{c \bar{p}(x)} dx \\ &= \int p(x) \log \frac{p(x)}{\bar{p}(x)} dx - \log(c) \int p(x) dx \\ &= D_{KL}(p \parallel \bar{p}) - \log(c). \end{aligned} \quad (\text{A.1})$$

Since the latter term $-\log c$ does not depend on p and the KLD of two PDFs is always non-negative and equal to zero if and only if the two PDFs coincide almost everywhere, minimization of $J(p)$ with respect to p is achieved by zeroing out the first term $D_{KL}(p \parallel \bar{p})$, i.e. by taking $p = \bar{p}$. Hence, the weighted KLA $\bar{p}(\cdot)$ defined via (3) coincides with the normalized (weighted) geometric mean given by (4). ■

Proof of Theorem 2. To prove the result, it is convenient to exploit the following elementary properties of Gaussian functions:

$$[\mathcal{N}(x; \mu, \Sigma)]^\pi \propto \mathcal{N}(x; \mu, \pi^{-1} \Sigma) \quad (\text{A.2})$$

$$\prod_{i=1}^N \mathcal{N}(x; \mu^i, \Sigma^i) \propto \mathcal{N}(x; \mu, \Sigma) \quad (\text{A.3})$$

where $\Sigma^{-1} = \sum_{i=1}^N (\Sigma^i)^{-1}$ and $\Sigma^{-1} \mu = \sum_{i=1}^N (\Sigma^i)^{-1} \mu^i$. The check of (A.2) is straightforward from the definition of Gaussian function; in fact

$$\begin{aligned} [\mathcal{N}(x; \mu, \Sigma)]^\pi &\propto \exp \left[-\frac{1}{2} (x - \mu)^\top [\pi^{-1} \Sigma]^{-1} (x - \mu) \right] \\ &\propto \mathcal{N}(x; \mu, \pi^{-1} \Sigma). \end{aligned}$$

To check (A.3), notice that

$$\begin{aligned} \prod_{i=1}^N \mathcal{N}(x; \mu^i, \Sigma^i) &\propto \exp \left\{ -\frac{1}{2} \left[x^\top \sum_{i=1}^N (\Sigma^i)^{-1} x - \left(\sum_{i=1}^N (\Sigma^i)^{-1} \mu^i \right)^\top x \right. \right. \\ &\quad \left. \left. - x^\top \sum_{i=1}^N (\Sigma^i)^{-1} \mu^i + \sum_{i=1}^N (\mu^i)^\top (\Sigma^i)^{-1} \mu^i \right] \right\}. \end{aligned}$$

Then, exploiting the relationships for Σ^{-1} and $\Sigma^{-1}\mu$ in (A.3), it is immediate to get

$$\begin{aligned} & \prod_{i=1}^N \mathcal{N}(x; \mu^i, \Sigma^i) \\ & \propto \exp \left\{ -\frac{1}{2} [x^\top \Sigma^{-1} x - \mu^\top \Sigma^{-1} x - x^\top \Sigma^{-1} \mu] \right\} \\ & \propto \exp \left[-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right] \propto \mathcal{N}(x; \mu, \Sigma). \end{aligned}$$

Now, assuming (10) and exploiting (A.2)–(A.3),

$$\begin{aligned} \bar{p}(x) & \propto \prod_{i=1}^N [\mathcal{N}(x; \mu^i, \Sigma^i)]^{\pi^i} \propto \prod_{i=1}^N \mathcal{N}\left(x; \mu^i, \frac{\Sigma^i}{\pi^i}\right) \\ & \propto \mathcal{N}(x; \bar{\mu}, \bar{\Sigma}) \end{aligned}$$

where $\bar{\mu}$ and $\bar{\Sigma}$ trivially satisfy (11) and (12). Hence, $\bar{p}(x) = c \mathcal{N}(x; \bar{\mu}, \bar{\Sigma})$ for some normalization constant $c \in \mathbb{R}$. Since

$$\int \bar{p}(x) dx = c \int \mathcal{N}(x; \bar{\mu}, \bar{\Sigma}) dx = c = 1$$

the result is proved. ■

For the sake of brevity, let us rewrite the prediction step (25) on the information matrix as

$$\Omega_{k+1|k}^i = \Psi(\Omega_{k|k}^i)$$

with

$$\Psi(\Omega) \triangleq A^{-\top} \Omega A^{-1} - A^{-\top} \Omega (\Omega + A^\top Q^{-1} A)^{-1} \Omega A^{-1}.$$

In order to prove Theorems 3 and 4, it is convenient to avail of the following technical Lemma.

Lemma 1. Under assumption A1 the following facts hold.

- (i) The function $\Psi(\cdot)$ is monotone nondecreasing, i.e., given two positive semidefinite matrices Ω_1 and Ω_2 with $\Omega_1 \leq \Omega_2$ one has $0 \leq \Psi(\Omega_1) \leq \Psi(\Omega_2)$.
- (ii) For any positive semidefinite matrix $\check{\Omega}$, there exists a strictly positive real $\check{\beta} \leq 1$ such that $\Psi(\Omega) \geq \check{\beta} A^{-\top} \Omega A^{-1}$ for any $\Omega \leq \check{\Omega}$.
- (iii) For any positive definite matrix $\tilde{\Omega}$, there exists a positive real $\tilde{\beta} < 1$ such that $\Psi(\Omega) \leq \tilde{\beta} A^{-\top} \Omega A^{-1}$ for any $\Omega \geq \tilde{\Omega}$.

Proof of Lemma 1. (i) Recall that, for a non-singular matrix Ω , one has

$$\Psi(\Omega) = (A\Omega^{-1}A^\top + Q)^{-1}. \quad (\text{A.4})$$

Consider now two matrices Ω_1 and Ω_2 with $\Omega_1 \leq \Omega_2$ and define $\Omega_1(\alpha) = \Omega_1 + \alpha I$, $\Omega_2(\alpha) = \Omega_2 + \alpha I$. Using (A.4), it is immediate to see that, for any $\alpha > 0$,

$$\Psi(\Omega_1(\alpha))^{-1} - \Psi(\Omega_2(\alpha))^{-1} = A(\Omega_1(\alpha)^{-1} - \Omega_2(\alpha)^{-1})A^\top \geq 0$$

and, hence, $\Psi(\Omega_2(\alpha)) - \Psi(\Omega_1(\alpha)) \geq 0$. Then the proof of fact (i) follows from continuity arguments by letting $\alpha \rightarrow 0^+$.

(ii) Consider a generic $\Omega \leq \check{\Omega}$ and let $\Omega(\alpha) = \Omega + \alpha I$ with $\alpha > 0$. Using again (A.4), we can write

$$\Psi(\Omega(\alpha)) = A^{-\top} (\Omega(\alpha)^{-1} + A^{-1}QA^{-\top})^{-1} A^{-1}. \quad (\text{A.5})$$

Notice now that, for any given $\check{\alpha}$, one has $\Omega(\alpha)^{-1} \geq (\check{\Omega} + \check{\alpha} I)^{-1}$ for any $0 < \alpha \leq \check{\alpha}$ and $0 \leq \Omega \leq \check{\Omega}$. This, in turn, implies that there exists a positive scalar $\check{\gamma}$ such that

$$A^{-1}QA^{-\top} \leq \check{\gamma} \Omega(\alpha)^{-1} \quad (\text{A.6})$$

for any $0 < \alpha \leq \check{\alpha}$ and $0 \leq \Omega \leq \check{\Omega}$. Combining (A.5) with (A.6), we get

$$\Psi(\Omega(\alpha)) \geq \frac{1}{1 + \check{\gamma}} A^{-\top} \Omega(\alpha) A^{-1}.$$

Then the proof of fact (ii) can be concluded by setting $\check{\beta} = 1/(1 + \check{\gamma})$ and noting that the latter inequality must hold also in the limit for $\alpha \rightarrow 0^+$.

(iii) Given a generic $\Omega \geq \tilde{\Omega} > 0$, we can write

$$\Psi(\Omega) = A^{-\top} (\Omega^{-1} + A^{-1}QA^{-\top})^{-1} A^{-1}.$$

Since $Q > 0$, there always exists a positive scalar $\tilde{\gamma}$ such that

$$A^{-1}QA^{-\top} \geq \tilde{\gamma} \tilde{\Omega}^{-1} \geq \tilde{\gamma} \Omega^{-1}$$

and, hence,

$$\Psi(\Omega) \leq \frac{1}{1 + \tilde{\gamma}} A^{-\top} \Omega A^{-1}.$$

Then the proof of fact (iii) can be concluded by setting $\tilde{\beta} = 1/(1 + \tilde{\gamma})$. ■

Proof of Theorem 3. It can be given by induction. To this end, suppose that at a generic time k

$$\Omega_{k|k-1}^i \leq (\mathbb{E} \{ (x_k - \hat{x}_{k|k-1}^i)(x_k - \hat{x}_{k|k-1}^i)^\top \})^{-1} \quad (\text{A.7})$$

for any $i \in \mathcal{N}$. Clearly, defining $\mathcal{I}^i \triangleq (C^i)^\top (R^i)^{-1} C^i$, (A.7) also implies

$$\begin{aligned} & (\mathbb{E} \{ (x_k - \hat{x}_{k|k,0}^i)(x_k - \hat{x}_{k|k,0}^i)^\top \})^{-1} \\ & = (\mathbb{E} \{ (x_k - \hat{x}_{k|k-1}^i)(x_k - \hat{x}_{k|k-1}^i)^\top \})^{-1} + \mathcal{I}^i \\ & \geq \Omega_{k|k-1}^i + \mathcal{I}^i = \Omega_{k|k,0}^i. \end{aligned}$$

Then, since the Covariance Intersection fusion rule preserves the consistency property (Chen et al., 2002; Julier & Uhlmann, 1997), i.e.

$$(E \{ \tilde{x}_{k|k,\ell}^i (\tilde{x}_{k|k,\ell}^i)^\top \})^{-1} \geq \Omega_{k|k,\ell}$$

implies

$$(E \{ \tilde{x}_{k|k,\ell+1}^i (\tilde{x}_{k|k,\ell+1}^i)^\top \})^{-1} \geq \Omega_{k|k,\ell+1}$$

for any $\ell = 0, \dots, L-1$, one can conclude that (31) holds at time k with $\hat{x}_{k|k}^i = \hat{x}_{k|k,L}^i$ and $\Omega_{k|k}^i = \Omega_{k|k,L}^i$. Further, in view of Lemma 1 fact (i), it is immediate to see that

$$\begin{aligned} \Omega_{k+1|k}^i & = \Psi(\Omega_{k|k}^i) \\ & \leq \Psi \left((\mathbb{E} \{ (x_k - \hat{x}_{k|k}^i)(x_k - \hat{x}_{k|k}^i)^\top \})^{-1} \right) \\ & = (E \{ (x_{k+1} - \hat{x}_{k+1|k}^i)(x_{k+1} - \hat{x}_{k+1|k}^i)^\top \})^{-1}. \end{aligned}$$

The proof can be concluded by noting that, with the initialization (28)–(29), inequality (A.7) automatically holds at $k = 0$. ■

Proof of Theorem 4. For the sake of brevity the proof is given for the case $L = 1$. The extension to a generic $L \geq 1$ is straightforward.

When $L = 1$, each local fused information matrix can be written as

$$\begin{aligned} \Omega_{k|k}^i & = \sum_{j \in \mathcal{N}^i} \pi^{ij} \Omega_{k|k-1}^j + \sum_{j \in \mathcal{N}^i} \pi^{ij} \mathcal{I}^j \\ & = \sum_{j \in \mathcal{N}^i} \pi^{ij} \Psi(\Omega_{k-1|k-1}^j) + \sum_{j \in \mathcal{N}^i} \pi^{ij} \mathcal{I}^j. \end{aligned} \quad (\text{A.8})$$

Further, in view of inequality (31), one can also claim that

$$\Omega_{k|k}^i \leq (\mathbb{E} \{ (x_k - \hat{x}_{k|k}^i)(x_k - \hat{x}_{k|k}^i)^\top \})^{-1} \leq \Omega_{k|k}$$

where $\Omega_{k|k}$ is the information matrix of the centralized Kalman filter (which, in this case, represents the Cramer–Rao upper bound on the information matrix achievable by any unbiased filter). Since Q and R^i , $i \in \mathcal{N}$, are positive definite, the sequence $\Omega_{k|k}$ can be uniformly bounded by some constant matrix $\tilde{\Omega}$. Thus, one can invoke Lemma 1 fact (ii) and obtain the lower bound

$$\Omega_{k|k}^i \geq \sum_{j \in \mathcal{N}^i} \pi^{ij} \tilde{\beta} A^{-\top} \Omega_{k-1|k-1}^j A^{-1} + \sum_{j \in \mathcal{N}^i} \pi^{ij} \mathcal{J}^j \quad (\text{A.9})$$

for some positive real $\tilde{\beta}$. By recursively applying Eqs. (A.8) and (A.9) a certain number, say $\bar{k}$, of times, it is possible to write

$$\begin{aligned} \Omega_{k|k}^i &\geq \tilde{\beta}^{\bar{k}} (A^{-\bar{k}})^\top \left(\sum_{j \in \mathcal{N}} \pi_{\bar{k}}^{ij} \Omega_{k-\bar{k}|k-\bar{k}}^j \right) A^{-\bar{k}} \\ &\quad + \sum_{\tau=1}^{\bar{k}} \tilde{\beta}^{\tau-1} (A^{-\tau+1})^\top \left(\sum_{j \in \mathcal{N}} \pi_{\tau}^{ij} \mathcal{J}^j \right) A^{-\tau+1} \end{aligned}$$

where π_{τ}^{ij} denotes the (i, j) th element of the matrix Π^τ . Then, the proof can be concluded by defining

$$\tilde{\Omega} \triangleq \sum_{\tau=1}^{\bar{k}} \tilde{\beta}^{\tau-1} (A^{-\tau+1})^\top \left(\sum_{j \in \mathcal{N}} \pi_{\tau}^{ij} \mathcal{J}^j \right) A^{-\tau+1}.$$

In fact, when the consensus matrix Π is primitive, the elements π_{τ}^{ij} are all positive for τ greater than a certain m . Then, it is an easy matter to verify that, under collective observability, the above-defined matrix $\tilde{\Omega}$ is positive definite for $\bar{k}$ greater than $m + n$. ■

The following technical Lemma will be useful in the proof of Theorem 5.

Lemma 2. Given an integer $N \geq 2$, N positive definite matrices $M_1, \dots, M_N$, and N vectors $v_1, \dots, v_N$, the following inequality holds

$$\left(\sum_{i=1}^N M_i v_i \right)^\top \left(\sum_{i=1}^N M_i \right)^{-1} \left(\sum_{i=1}^N M_i v_i \right) \leq \sum_{i=1}^N v_i^\top M_i v_i. \quad (\text{A.10})$$

Proof of Lemma 2. The proof is given by induction. Consider first the case $N = 2$. Then, the left-hand side of (A.10) becomes

$$\begin{aligned} &(M_1 v_1 + M_2 v_2)^\top (M_1 + M_2)^{-1} (M_1 v_1 + M_2 v_2) \\ &= v_1^\top M_1 (M_1 + M_2)^{-1} M_1 v_1 + v_2^\top M_2 (M_1 + M_2)^{-1} M_2 v_2 \\ &\quad + v_1^\top M_1 (M_1 + M_2)^{-1} M_2 v_2 + v_2^\top M_2 (M_1 + M_2)^{-1} M_1 v_1. \end{aligned}$$

Since the matrix $M_1(M_1 + M_2)^{-1}M_2 = (M_1^{-1} + M_2^{-1})^{-1} = M_2(M_1 + M_2)^{-1}M_1$ is positive definite, one has

$$\begin{aligned} &v_1^\top M_1 (M_1 + M_2)^{-1} M_2 v_2 + v_2^\top M_2 (M_1 + M_2)^{-1} M_1 v_1 \\ &\leq v_1^\top M_1 (M_1 + M_2)^{-1} M_2 v_1 + v_2^\top M_2 (M_1 + M_2)^{-1} M_1 v_2 \end{aligned}$$

and hence

$$\begin{aligned} &(M_1 v_1 + M_2 v_2)^\top (M_1 + M_2)^{-1} (M_1 v_1 + M_2 v_2) \\ &\leq v_1^\top M_1 v_1 + v_2^\top M_2 v_2 \end{aligned} \quad (\text{A.11})$$

which proves the claim for $N = 2$. Consider now a generic N and suppose that (A.10) holds for $N - 1$, i.e.,

$$\left(\sum_{i=1}^{N-1} M_i v_i \right)^\top \left(\sum_{i=1}^{N-1} M_i \right)^{-1} \left(\sum_{i=1}^{N-1} M_i v_i \right) \leq \sum_{i=1}^{N-1} v_i^\top M_i v_i. \quad (\text{A.12})$$

By defining

$$M = \sum_{i=1}^{N-1} M_i, \quad v = M^{-1} \left(\sum_{i=1}^{N-1} M_i v_i \right),$$

we can write

$$\begin{aligned} &\left(\sum_{i=1}^N M_i v_i \right)^\top \left(\sum_{i=1}^N M_i \right)^{-1} \left(\sum_{i=1}^N M_i v_i \right) \\ &= (M v + M_N v_N)^\top (M + M_N)^{-1} (M v + M_N v_N) \\ &\leq v^\top M v + v_N^\top M_N v_N \end{aligned}$$

where the latter inequality follows from (A.11). The proof is concluded by noting that

$$v^\top M v = \left(\sum_{i=1}^{N-1} M_i v_i \right)^\top \left(\sum_{i=1}^{N-1} M_i \right)^{-1} \left(\sum_{i=1}^{N-1} M_i v_i \right)$$

and, then, by applying (A.12). ■

Proof of Theorem 5. For the sake of compactness, let us denote the prediction and estimation error at node i by $\tilde{x}_{k|k-1}^i \triangleq x_k - \hat{x}_{k|k-1}^i$ and $\tilde{x}_{k|k}^i \triangleq x_k - \hat{x}_{k|k}^i$, respectively. In view of Theorem 4 and Lemma 1 fact (iii), one has

$$\begin{aligned} \mathcal{L}_{k+1}^i(\tilde{x}_{k+1|k}^i) &= (\tilde{x}_{k+1|k}^i)^\top \Omega_{k+1|k}^i \tilde{x}_{k+1|k}^i \\ &= [(\tilde{x}_{k+1|k}^i)^\top \Psi (\Omega_{k|k}^i \tilde{x}_{k+1|k}^i)] \\ &\leq \tilde{\beta} [(\tilde{x}_{k+1|k}^i)^\top A^{-\top} \Omega_{k|k}^i A^{-1} \tilde{x}_{k+1|k}^i] \end{aligned}$$

for some positive real $\tilde{\beta} < 1$ and for $k \geq \bar{k}$. This, in turn, implies

$$\mathcal{L}_{k+1}^i(\tilde{x}_{k+1|k}^i) \leq \tilde{\beta} [(\tilde{x}_{k|k}^i)^\top \Omega_{k|k}^i \tilde{x}_{k|k}^i] \quad (\text{A.13})$$

since $\tilde{x}_{k+1|k}^i = A \tilde{x}_{k|k}^i$.

Notice now that the noise-free local correction step of Algorithm 2 can be written as

$$\Omega_{k|k,0}^j \hat{x}_{k|k,0}^j = \Omega_{k|k-1}^j \hat{x}_{k|k-1}^j + \mathcal{J}^j x_k \quad (\text{A.14})$$

where $\mathcal{J}^j \triangleq (C^j)^\top (R^j)^{-1} C^j$. The above expression has been obtained from (22) with $y_k^j = C^j x_k$ in place of y_k^j , $q_{k|k,0}^j = \Omega_{k|k,0}^j \hat{x}_{k|k,0}^j$ in place of $q_{k|k}^j$, and $q_{k|k-1}^j = \Omega_{k|k-1}^j \hat{x}_{k|k-1}^j$ in place of $q_{k|k-1}^j$. Recall also that, by consensus,

$$\begin{aligned} \Omega_{k|k}^i &\triangleq \Omega_{k|k,L}^i = \sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k,0}^j \\ \Omega_{k|k}^i \hat{x}_{k|k}^i &\triangleq \Omega_{k|k,L}^i \hat{x}_{k|k,L}^i = \sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k,0}^j \hat{x}_{k|k,0}^j. \end{aligned} \quad (\text{A.15})$$

Hence, pre-multiplying the first equation of (A.15) by $(\Omega_{k|k}^i)^{-1}$ and post-multiplying it by x_k , one gets

$$x_k = (\Omega_{k|k}^i)^{-1} \sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k,0}^j x_k. \quad (\text{A.16})$$

In a similar way, pre-multiplying the second equation of (A.15) by $(\Omega_{k|k}^i)^{-1}$, one gets

$$\hat{x}_{k|k}^i = (\Omega_{k|k}^i)^{-1} \sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k,0}^j \hat{x}_{k|k,0}^j. \quad (\text{A.17})$$

Next, subtracting (A.17) from (A.16), one obtains

$$\begin{aligned} \tilde{x}_{k|k}^i &\triangleq x_k - \hat{x}_{k|k}^i \\ &= (\Omega_{k|k}^i)^{-1} \sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k,0}^j (x_k - \hat{x}_{k|k,0}^j). \end{aligned}$$

Since in the local correction step $\Omega_{k|k,0}^j = \Omega_{k|k-1}^j + \mathcal{L}^j$, one has also

$$\Omega_{k|k,0}^j x_k = \Omega_{k|k-1}^j x_k + \mathcal{L}^j x_k. \quad (\text{A.18})$$

Then, exploiting (A.14) and (A.18) in (A.17), the estimation error can be rewritten as

$$\tilde{x}_{k|k}^i = (\Omega_{k|k}^i)^{-1} \sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k-1}^j \tilde{x}_{k|k-1}^j.$$

Exploiting the above identity, (A.13) can be rewritten as

$$\begin{aligned} \mathcal{L}_{k+1}^i(\tilde{x}_{k+1|k}^i) &\leq \tilde{\beta} \left[\left(\sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k-1}^j \tilde{x}_{k|k-1}^j \right)^\top \right. \\ &\quad \times \left. (\Omega_{k|k}^i)^{-1} \left(\sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k-1}^j \tilde{x}_{k|k-1}^j \right) \right]. \quad (\text{A.19}) \end{aligned}$$

Since $\Omega_{k|k}^i \geq \sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k-1}^j$, (A.19) implies also

$$\begin{aligned} \mathcal{L}_{k+1}^i(\tilde{x}_{k+1|k}^i) &\leq \tilde{\beta} \left[\left(\sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k-1}^j \tilde{x}_{k|k-1}^j \right)^\top \left(\sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k-1}^j \right)^{-1} \right. \\ &\quad \times \left. \left(\sum_{j \in \mathcal{N}} \pi_L^{ij} \Omega_{k|k-1}^j \tilde{x}_{k|k-1}^j \right) \right]. \quad (\text{A.20}) \end{aligned}$$

Finally, by applying Lemma 2 to the right-hand side of (A.20), it can be seen that

$$\begin{aligned} \mathcal{L}_{k+1}^i(\tilde{x}_{k+1|k}^i) &\leq \tilde{\beta} \sum_{j \in \mathcal{N}} \pi_L^{ij} (\tilde{x}_{k|k-1}^j)^\top \Omega_{k|k-1}^j \tilde{x}_{k|k-1}^j \\ &= \tilde{\beta} \sum_{j \in \mathcal{N}} \pi_L^{ij} \mathcal{L}_k^j(\tilde{x}_{k|k-1}^j). \quad \blacksquare \end{aligned}$$

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